KINGDOM OF SAUDI ARABIA

Taibah University

College of Computer Science and Engineering

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AI201: Fundamentals of Artificial Intelligence

Laptop Recommendation Expert System

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Section 1: Project Introduction and Description

1.1 Introduction

In this project, we are going to build a Laptop Recommendation Expert System using Prolog.

The main idea of the project is to help users choose a suitable laptop based on their needs and preferences.

Choosing a laptop can be confusing because there are many laptop types, prices, and specifications. Some laptops are better for studying, some are better for programming, while others are more suitable for gaming, business, or design. Because of that, the user may not always know which laptop is the best choice.

Our system will make this process easier by asking the user about five important factors: budget, usage, battery life, portability, and performance.

Then, the system will use Prolog facts and rules to give a suitable recommendation.

This project is related to Artificial Intelligence because it uses the idea of an expert system, where the system tries to make a decision based on stored knowledge and logical rules.

1.2 Problem Description

Many people face difficulty when choosing a laptop because they may not understand all the technical specifications. For example, a student may need a laptop with long battery life and high portability, while a gamer may need a laptop with high performance. Also, a designer may need a strong laptop that can handle design software, while a business user may need a practical and lightweight laptop.

The problem is that users may choose a laptop based only on price or appearance without checking if it really matches their needs. This can lead to buying a laptop that is not suitable, too expensive, or not powerful enough for the required tasks.

For this reason, our project aims to provide a simple system that helps the user make a better decision. Instead of comparing many laptops manually, the user enters their preferences, and the system recommends a laptop category based on clear rules.

1.3 Expert System Idea

An expert system is a type of Artificial Intelligence system that is designed to solve problems in a specific area by using knowledge and rules. It works in a similar way to how a human expert gives advice.

In our project, the expert system works like a laptop advisor. It uses information about laptop features and matches them with the user's requirements. The system does not guess randomly.

Instead, it follows rules that are already written in the knowledge base.

For example, if the user wants a laptop for gaming, has a high budget, and needs high performance, the system can recommend a gaming laptop. If another user wants a laptop for study with long battery life and high portability, the system can recommend a study laptop.

The expert system in this project depends on the following attributes:

Attribute	**Values**
Budget	low, medium, high
Usage	basic, student, gaming, business, design
Battery	short, medium, long
Portability	low, medium, high
Performance	basic, medium, high

1.4 Why Prolog is Used

Prolog is used in this project because it is a logic programming language. It is suitable for expert systems because it works with facts, rules, and queries.

In Prolog, we can write facts to describe the available laptop options and their features. Then, we can write rules that connect these features with the correct recommendation. After that, the user can ask the system a query, and Prolog will search for the answer based on the facts and rules.

This makes Prolog useful for our project because the laptop recommendation depends on logical conditions. For example, the system can check if the user's budget, usage, battery life, portability, and performance match a certain laptop type.

1.5 How Prolog Uses Facts and Rules in This Project

In this project, facts are used to store information about laptop models and their features. A fact can describe one laptop using the selected attributes.

For example:

feature(macbook_air, high, student, long, high, medium).

This fact means that the MacBook Air has:

- high budget
- student usage

- long battery life
- high portability
- medium performance

Rules are used to make the recommendation. A rule checks the user's input and decides which laptop is suitable.

For example:

```
recommend_student_laptop(Laptop) :-
  feature(Laptop, _, student, long, high, _).
```

This rule means that if the user wants a laptop for a student with long battery life and high portability, the system will recommend a suitable student laptop from the knowledge base.

So, the system works by comparing the user's answers with the facts and rules. If the conditions match, Prolog returns the suitable recommendation. This is how the expert system uses logical reasoning to help the user choose a laptop.

1.6 Project Scope

The scope of this project is to build a simple laptop recommendation expert system using Prolog.

The system will recommend laptops based only on five attributes: budget, usage, battery life, portability, and performance.

The system will not compare real laptop prices from online stores, and it will not check laptop availability. It will only depend on the facts and rules written in the Prolog knowledge base.

This scope is suitable for our project because the main goal is to understand how expert systems work and how Prolog can be used to make decisions based on logic.

Section 2: Knowledge Base

2.1 Overview of The Knowledge Base:

The Knowledge Base is the main component of the Laptop Recommendation Expert System. It stores the facts and logical rules used by the system to recommend laptops based on user needs. The knowledge base contains information about laptop models and their features, including budget level, usage type, battery life, portability, and performance level. These facts and rules help the system generate recommendations using Prolog logic.

2.2 Knowledge Representation Categories:

The expert system represents laptop information using the five attributes mentioned in Section 1.3 **(Budget, Usage, Battery, Portability, and Performance)**. Each laptop is represented in Prolog using the feature/6 predicate, where the laptop name is followed by its five attribute values.

The general structure of the predicate is:

prolog

feature(LaptopName, Budget, Usage, Battery, Portability, Performance).

This structure ensures that all laptops in the knowledge base follow a consistent format, making it easier for the expert system to compare features and generate recommendations.

2.3 Laptop Feature Facts:

The following table shows the laptop models used in the system and their main features:

Laptop Model	Budget	Usage	Battery	Portability	Performance
MacBook Air	high	student	long	high	medium
MacBook Pro	high	design	long	medium	high
Dell XPS 13	high	business	long	high	medium
Dell XPS 15	high	design	medium	medium	high
Lenovo ThinkPad E14	medium	student	medium	medium	medium
HP Pavilion 15	low	basic	medium	low	basic
ASUS ROG Strix	high	gaming	short	low	high
MSI Gaming GF63	medium	gaming	short	medium	high
Surface Laptop 5	high	business	long	high	medium
Acer Aspire 5	low	basic	medium	medium	basic

1. **feature(macbook_air, high, student, long, high, medium).**
2. **feature(macbook_pro, high, design, long, medium, high).**
3. **feature(dell_xps_13, high, business, long, high, medium).**
4. **feature(dell_xps_15, high, design, medium, medium, high).**
5. **feature(lenovo_thinkpad_e14, medium, student, medium, medium, medium).**
6. **feature(hp_pavilion_15, low, basic, medium, low, basic).**
7. **feature(asus_rog_strix, high, gaming, short, low, high).**
8. **feature(msi_gaming_gf63, medium, gaming, short, medium, high).**

G. **feature(surface_laptop_5, high, business, long, high, medium).**

10. **feature(acer_aspire_5, low, basic, medium, medium, basic).**

2.4 Knowledge Base Rules:

The expert system uses logical rules to analyze laptop features and generate recommendations based on user requirements.

recommend_student_laptop(Laptop) :-

feature(Laptop, _, student, long, high, _).

recommend_gaming_laptop(Laptop) :-

feature(Laptop, _, gaming, _, _, high).

recommend_business_laptop(Laptop) :-

feature(Laptop, _, business, long, high, _).

2.5 Explanation of the Knowledge Representation:

The system represents laptop information using the feature/6 predicate in Prolog. Each fact contains the laptop model and its main features, **such as budget level, usage type, battery life, portability, and performance level.**

This structure helps the system organize and compare laptop features more easily when recommending laptops to users.

For example:

feature(macbook_air, high, student, long, high, medium).

This fact means that the **MacBook Air** is a high-budget laptop suitable for students, with long battery life, high portability, and medium performance.

2.6 Conclusion:

The knowledge base helps the expert system store and compare laptop information in an organized way. Using Prolog facts and rules allows the system to recommend laptops based on the user's needs.

Section 3: Codes

3.1 Rules and Logic

The rule section connects user requirements with laptop features stored in the knowledge base.

The Expert System uses Prolog rules to analyze the laptop specifications and provide suitable recommendations based on budget, usage, battery life, portability, and performance.

3.2 Prolog Rules

```
% Student Laptop
recommend_student_laptop(Laptop) :-
    feature(Laptop, _, student, _, _, _).

% Gaming Laptop
recommend_gaming_laptop(Laptop) :-
    feature(Laptop, _, gaming, _, _, high).

% Business Laptop
recommend_business_laptop(Laptop) :-
    feature(Laptop, _, business, long, high, _).

% Design Laptop
recommend_design_laptop(Laptop) :-
    feature(Laptop, _, design, _, _, high).

% Basic Laptop
recommend_basic_laptop(Laptop) :-
    feature(Laptop, low, basic, _, _, _).

% Long battery Laptop
recommend_long_battery_laptop(Laptop) :-
    feature(Laptop, _, _, long, _, _).
```

```prolog
% Portable laptop
recommend_portable_laptop(Laptop) :-
    feature(Laptop, _, _, _, high, _).

% High performance laptop
recommend_high_performance_laptop(Laptop) :-
    feature(Laptop, _, _, _, _, high).

% Low budget laptop
recommend_low_budget_laptop(Laptop) :-
    feature(Laptop, low, _, _, _, _).

% Medium budget laptop
recommend_medium_budget_laptop(Laptop) :-
    feature(Laptop, medium, _, _, _, _).

% High budget laptop
recommend_high_budget_laptop(Laptop) :-
    feature(Laptop, high, _, _, _, _).

% Gaming with high budget
recommend_premium_gaming_laptop(Laptop) :-
    feature(Laptop, high, gaming, _, _, high).
```

```prolog
% Student with long battery
recommend_best_student_laptop(Laptop) :-
    feature(Laptop, _, student, long, high, _).

% Business with portability
recommend_business_travel_laptop(Laptop) :-
    feature(Laptop, _, business, long, high, _).

% Main recommendation rule
recommend_laptop(Laptop, Budget, Usage, Battery, Portability, Performance) :-
    feature(Laptop, Budget, Usage, Battery, Portability, Performance).
```

3.3 Main Recommendation Rule

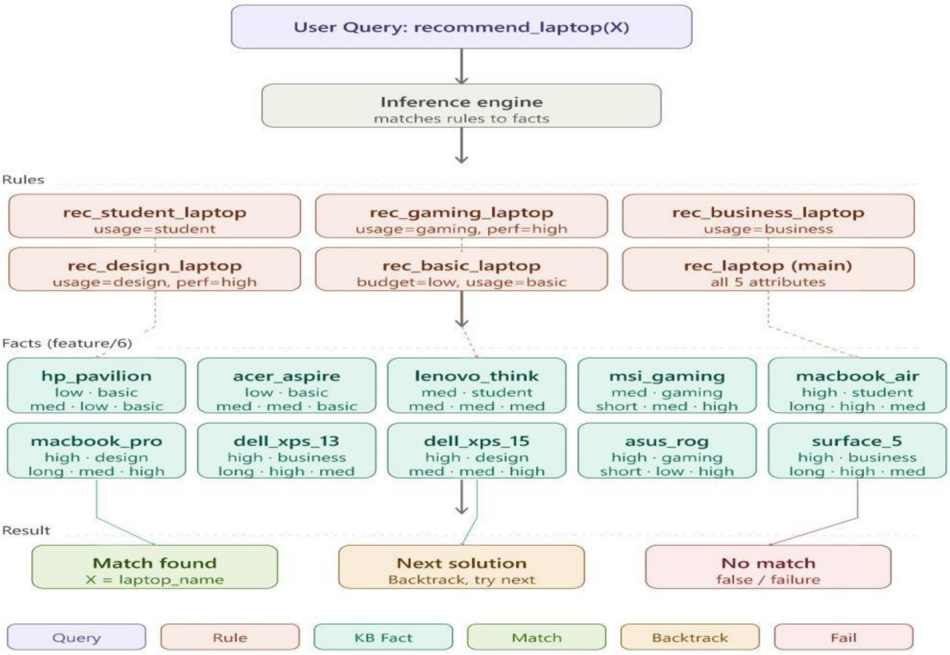
The main recommendation rule allows the system to recommend laptops based on multiple user requirements at the same time.

The system compares the user inputs with the laptop features stored in the knowledge base.

```prolog
% Main recommendation rule
recommend_laptop(Laptop, Budget, Usage, Battery, Portability, Performance) :-
    feature(Laptop, Budget, Usage, Battery, Portability, Performance).
```

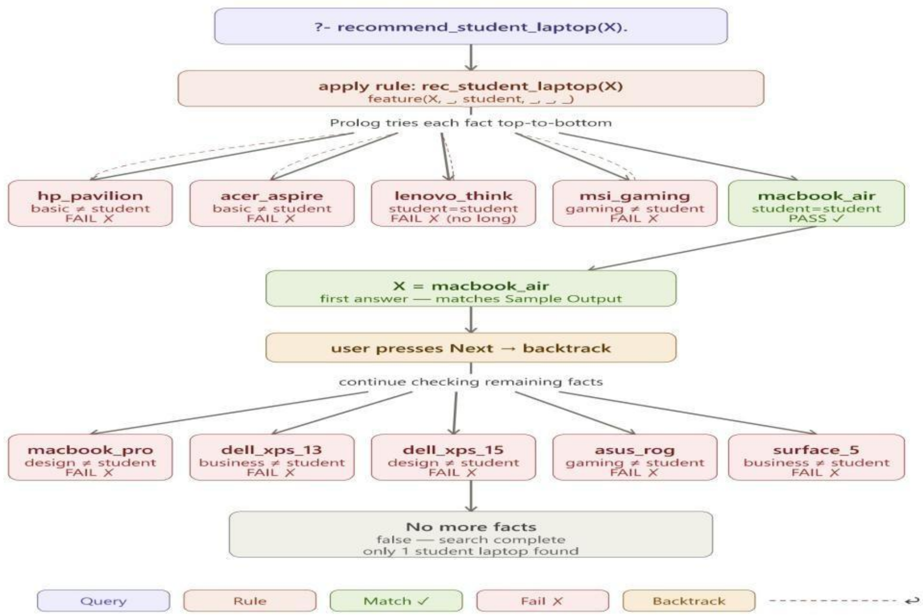
Section 4: Search Tree + Sample outputs

4.1 Search Tree



Search Tree 1 — Overview:

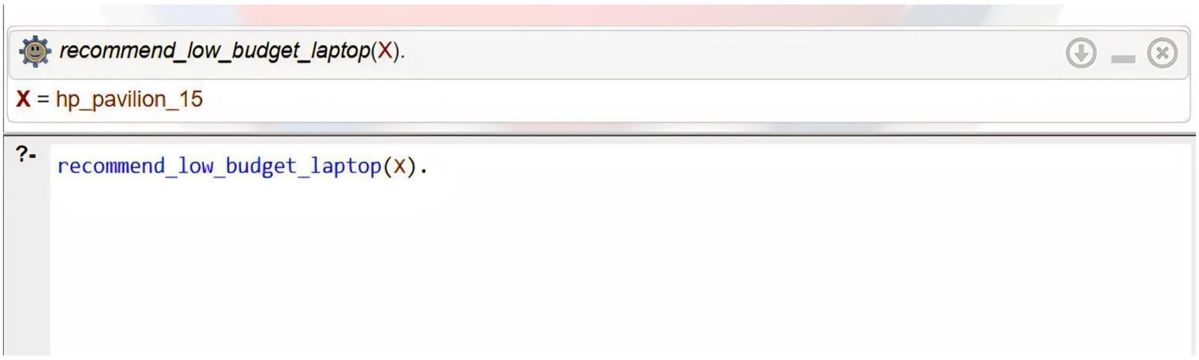
This search tree shows the full structure of the Laptop Recommendation Expert System. It illustrates how the inference engine connects the user query to the rules, and how each rule searches through the 10 laptop facts stored in the knowledge base. The result can be a match, a backtrack to find more solutions, or a failure if no facts satisfy the conditions.

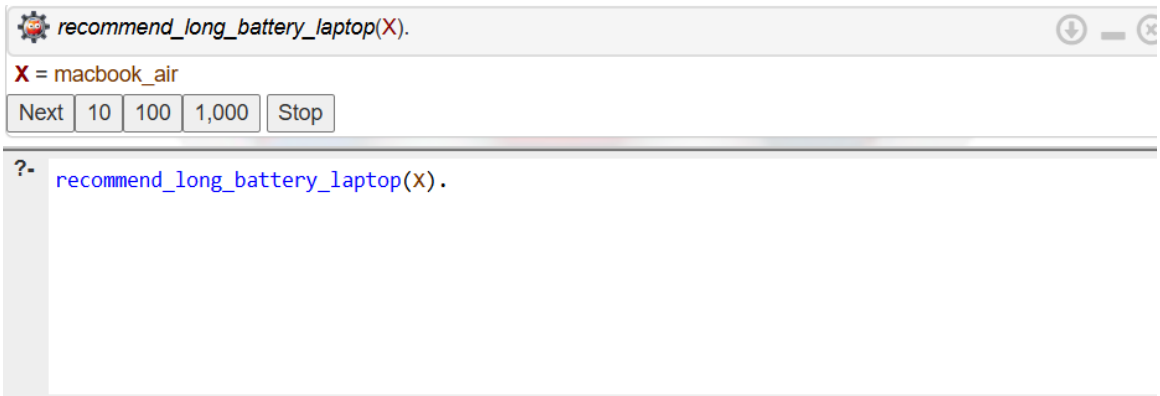


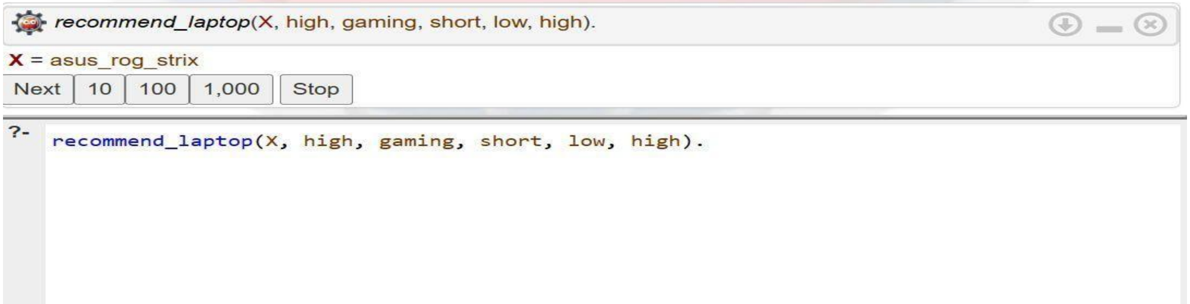
Search Tree 2 — recommend_student_laptop(X):

This search tree shows a detailed example of how Prolog processes the query recommend_student_laptop(X). Prolog applies the rule and checks each fact from top to bottom. Facts that do not match the condition (usage = student) are rejected and Prolog backtracks to try the next fact. The only matching fact is macbook_air, which is returned as the answer. After pressing Next, Prolog continues searching the remaining facts but finds no further matches, so the search ends with false.

4.2 Sample Outputs







RESOURCES:

[1] P. Blackburn, J. Bos, and K. Striegnitz, "Chapter 1: Facts, Rules, and Queries," in *Learn Prolog Now!*, SWI-Prolog ed., 2012. [Online]. Available: https://lpn.swi-prolog.org/lpnpage.php?pageid=lpn-htmlch1&pagetype=html. [Accessed: 11-May-2026].

[2] J. Dukes, "Backtracking in PROLOG (with CODE)," *YouTube*, May 2021. [Online]. Available: https://www.youtube.com/watch?v=PK2jQesRzgY. [Accessed: 14-May-2026].